Quantum Machine Learning Symmetry Score Solution

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1. Problem Overview & Symmetry Definition

We are given binary strings $x \in \mathbb{F}_2^n$ with class labels $f(x)$. We evaluate how compatible a candidate symmetry vector $s \in \mathbb{F}_2^n$ ($s \neq 0$) is with the observed dataset. A symmetry implies that labels are invariant under XOR translation:

```
f(x) = f(x ⊕ s)
```

Thus, x and $x \oplus s$ belong to the same equivalence class and share identical labels.

2. Quantum State Preparation

For each distinct class label r, we collect all training inputs belonging to that class into set $X_r = \{x_i : f(x_i) = r\}$ with total samples $N = \sum |X_r|$. We encode these inputs into a quantum state:

```
|X_r⟩ = (1 / √(|X_r|)) ∑_{x ∈ X_r} |x⟩
```

Applying Hadamard gates to every qubit ($H^{\otimes n}$) transforms the state. When X_r contains complete pairs $\{x, x \oplus s\}$, surviving superposition states satisfy:

```
H_s^⊥ = {y : y · s = 0 (mod 2)}
```

3. Ancilla Qubit Circuit & Probability $p_r(s)$

An ancilla qubit initialized in $|0\rangle$ interacts via CNOT gates controlled by data qubits where $s_j = 1$. The probability of measuring the ancilla in $|0\rangle$ measures how strongly class r respects symmetry s:

$$
\begin{aligned}
&p_r(s) = P(y \cdot s = 0) = \\
&\bullet\ 1.0 \quad \text{(if } X_r \text{ consists entirely of complete pairs } \{x,\ x \oplus s\}) \\
&\bullet\ 0.5 \quad \text{(if } X_r \text{ contains unpaired points or singletons)}
\end{aligned}
$$

4. Overall Symmetry Score Calculation

The final weighted symmetry score $S(s)$ combines class probabilities proportional to their relative sizes:

$$S(s) = \sum_r \left(|X_r| / N\right) \cdot p_r(s)$$

5. Summary Table of Public Test Cases

Test Case	Candidate Symmetry (s)	Class Structure & Analysis	Expected Score S(s)
1	[1, 1, 0]	Classes 0 & 1 form complete pairs; Class 2 is a single point.	**0.9**
2	[1, 1, 0]	Class 0 forms complete pairs {000, 110} and singleton {001}.	**0.8333...**

Analytical Implementation Note: By evaluating XOR pairs directly in Python, we bypass complex quantum gate simulation while perfectly matching the measurement statistics defined by the Hidden Subgroup Problem framework.